

$$\begin{array}{lcl}
 |\downarrow\rangle & & f_{\uparrow}^{\dagger} |0\rangle = |\uparrow\rangle \\
 & \rightarrow & f_{\downarrow}^{\dagger} |0\rangle = |\downarrow\rangle \\
 |\uparrow\rangle & & f_{\uparrow}^{\dagger} f_{\downarrow}^{\dagger} |0\rangle = |\uparrow\downarrow\rangle \\
 & & |0\rangle = |\emptyset\rangle
 \end{array}
 \left. \begin{array}{l} \\ \\ \\ \end{array} \right\} \begin{array}{l} \mathcal{H}_{\text{phys}} \\ \\ \mathcal{H}_{\text{aux}} \end{array}$$